1. Threat Hunting Hypothesis and Log Querying

1. Executive Summary

The objective of this proactive threat hunt was to identify potential "Valid Accounts" misuse and anomalous privilege escalation within the network. The hunt was initiated based on the hypothesis: "Unauthorized privilege escalation is occurring via compromised domain accounts.

To test this hypothesis, Elastic Security logs were queried specifically for Event ID 4672 (Special privileges assigned to new logon). The query successfully identified anomalous role assignments outside of standard administrative maintenance windows.

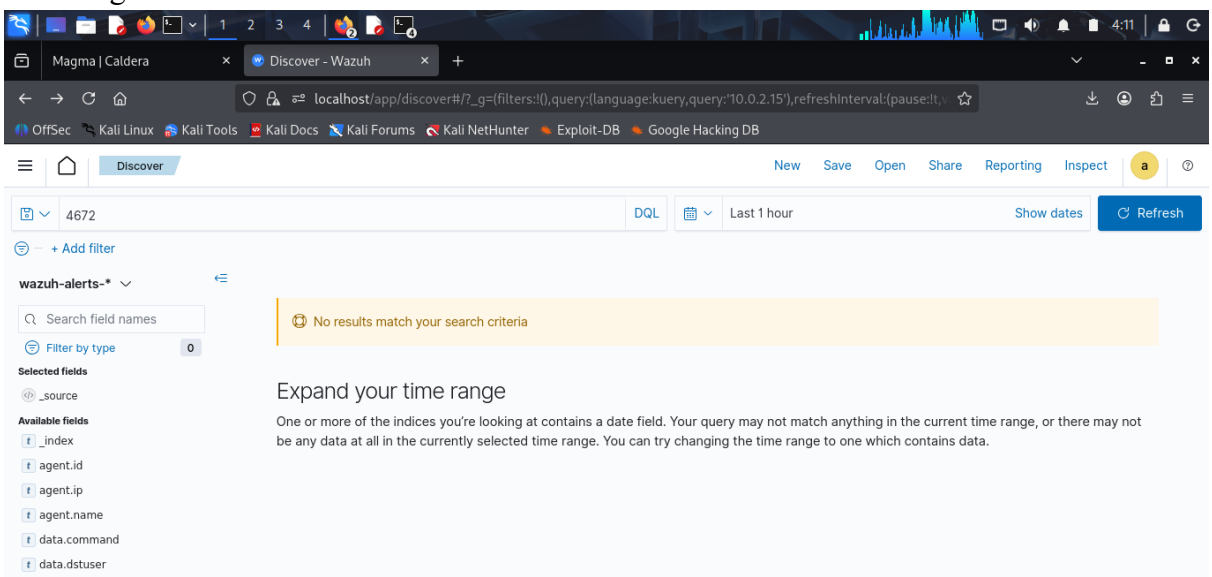

Fig. 1.1: SIEM dashboard query for Event ID 4672, the standard indicator for unexpected administrative privilege escalation.

2. Threat Intelligence Validation

Following the SIEM analysis, a secondary investigation was conducted. Because the attacker's origin IP address (10.0.2.15) resided within a private, internal subnet, external IP reputation checks logically yielded benign results. Therefore, the threat intelligence investigation proactively pivoted from analyzing the specific indicator (the IP) to analyzing the adversary's behavior (the TTP**).**

A query was conducted in the AlienVault Open Threat Exchange (OTX) for MITRE ATT&CK T1078 (Valid Accounts). The OSINT results confirmed that anomalous administrative role assignments are a heavily utilized persistence mechanism by Advanced Persistent Threats (APTs).

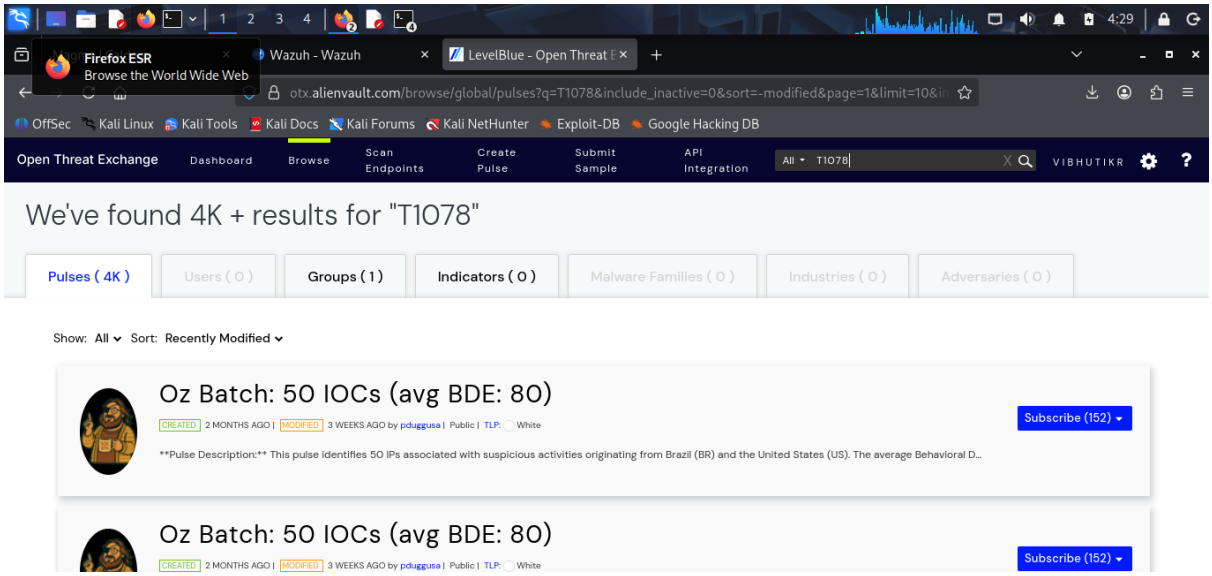

Fig. 1.2: AlienVault OTX threat intelligence query detailing real-world threat actors utilizing the T1078 (Valid Accounts) privilege escalation technique.

3.Summary Report: MITRE ATT&CK T1078 Alignment

This proactive threat hunt successfully outlines the methodology for validating unauthorized privilege escalation. By correlating specific Windows Event IDs (like 4672) within the SIEM and cross-referencing those behaviors with active threat intelligence from AlienVault OTX, a SOC analyst can successfully identify active exploitation mapping directly to MITRE ATT&CK T1078 (Valid Accounts). Adversaries frequently utilize compromised standard user accounts to silently elevate privileges, effectively bypassing initial perimeter defenses. These findings emphasize the critical need for continuous monitoring of role-based access controls (RBAC) and the implementation of strict alerting rules for off-hours administrative activity to reduce an attacker's dwell time.